

Comparison between Terrestrial and Celestial Communication

Communication parameters	Wired terrestrial	Mobile Ad Hoc NET/ MANET/ Wireless	IPN/Celestial/Wireless
Power availability	Not critical	Important	Very crucial
Signal-to-noise ration	Within acceptable range	Low	Very low
Error rate	Within acceptable range	Medium	High
Infrastructure	Defined/fixed	Deployable	Deployable
Medium	Copper/fibre	RF/IR	Primarily free space, RF
Delay in seconds	<1		10 to 10,000 (10,000 seconds means about three hours)
Deployment cost	Low	Medium	Very high
Operational cost	Low	Medium	Very high
Repairing and upgradation	Low	Medium	Very high

provide excellent opportunities to service providers and customers. IPTV offers unique opportunities to integrate voice, video and data over a broadband network. It is a simple, low-cost system of delivering TV services using IP to transport audio/video signals.

Use of IP for carrying audio, video and image signals is not new. What is new is the use of broadband consumer bandwidth and IP to meet desired performance and QoS for providing TV services. It is a form of commercial broadcasting TV using IPs, including IP, Transmission Control Protocol/User Datagram Protocol (TCP/UDP) and Real-Time Protocol/Real-Time Control Protocol (RTP/RTCP).

The carrier network is provided by service providers. IPTV, simply speaking, is a system of audio and video delivery over an Internet connection. The difference between IPTV and Internet TV is that, IPTV uses a dedicated network for distribution like very high rate digital subscriber line (VDSL) for broadband support, usually 25mbps, whereas Internet TV uses the whole infrastructure or the whole Internet as the transport medium.

IPTV has limited scope for users as it is considered to exist in a closed network. While, Internet TV is available and accessible everywhere, subject to Internet access availability.

Interplanetary Internet

Everywhere (terrestrial and celestial) Internet access and availability is the motto of Interplanetary Internet (IPN). Internet over the years has

undergone far-reaching and profound changes while becoming a critical communications infrastructure. It has become a fully-pervasive infrastructure providing anywhere, anytime and any service connectivity.

For the future Internet, the most important challenge is to ensure anywhere, everywhere communication support in the entire solar system. In data communication, anywhere and everywhere IP is the next-generation Internet technology.

IPN is next to IPv6, IPTV, VoIP, compressed header IP, mobile IP and other IPng (IP next generation). It is, in fact, an attempt of human ingenuity to move towards the future integration of space and terrestrial communication that will support the migration of human intelligence throughout the solar system.

IPN's vision is to provide Internet-like services across the solar system. IPN will look like the Internet on Earth and a similar one on Mars linked by gateways. These gateways are satellites that will relay the information between the Internets. How?

Like terrestrial or classical Internet, IPN will support:

- Best effort delivery of packets over stateless infrastructure
- Unique global addressing with dynamic routing
- End-to-end reliable service and flow control
- Scalable global domain names decoupled from addressing

IPN will not be like the Internet because of the differences in communication parameters between

IPN and classical Internet (see table above). Main differences imply that IPN protocol cannot rely on end-to-end connectivity; here, end-to-end delay is huge and cost of transmission is high.

This implies designing IPN as a network of regional internets spanning dissimilar environments and as a delay-tolerant network is important. In IPN what is needed is to transfer packets end-to-end through disconnected multiple regions that tolerate variable delay.

A protocol of IPN takes care of these requirements using Bundling Protocol Suite, which is believed to provide general-purpose delay tolerant protocol. IPN proposes to interconnect IPN regions by way of common messaging layers called bundles.

Bundling is a new overlay protocol to put together a set of heterogeneous Internets. Bundles (messages) are arbitrarily long messages designed for end-to-end delivery between IPN nodes over distinct or identical transport layers. These are routed by a routing function through a concatenated series of Internets, like how terrestrial IP routes packets through a series of independent sub-networks.

To achieve guaranteed end-to-end delivery, bundles may be retransmitted as in Automatic Repeat Request (ARQ) protocol used in TCP layer of terrestrial Internet. The interplanetary backbone of IPN depends on very fragile wireless links. Besides, hubs on the interplanetary backbone move with respect to each other. This may disturb the line-of-sight communication, which needs some sort of relay communication from time to time. Hence, unlike Earth's backbone environment of continuous connectivity, IPN transmission may get discontinued.

Bundles are routed, forwarded and custody-transferred with new concepts. In IPN, routing means sending using best efforts to the next IPN hop for the desired destination. Forwarding means sending bundles on demand for non-persistent nodes. Custody transfer is a reliable intra-IPN delivery with storage for persistent nodes. In fact, in custodial